

International Spent Nuclear Fuel Options

ENG 471 Seminar

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<http://katyhuff.github.io/2017-09-27-ne471.pdf>

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I L L I N O I S



Interactive Slides

The web-interactive first half of the talk can be found at:

<https://kathyhuff.github.io/2017-09-27-eng471>



Outline

① Introduction

- Nuclear Nations
- Spent Fuel Inventory

② Spent Fuel Options

- Long-Term Storage
- Reprocessing
- Deep Geologic Disposal



Nuclear Power Nations

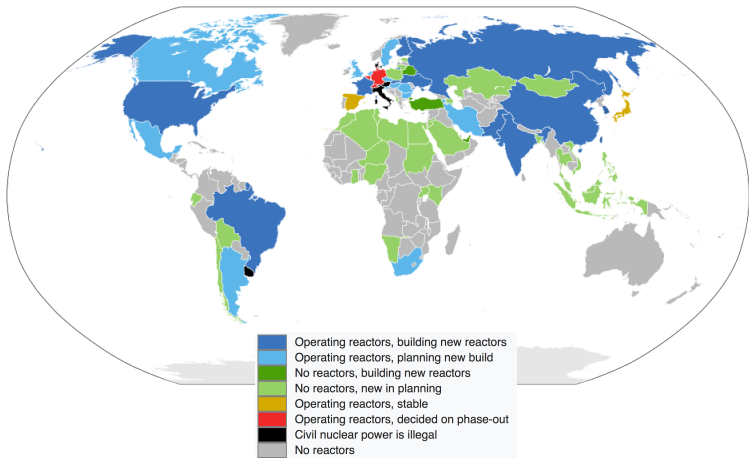


Figure 1: Nuclear power status of all nations [12].



Nuclear Weapons Nations

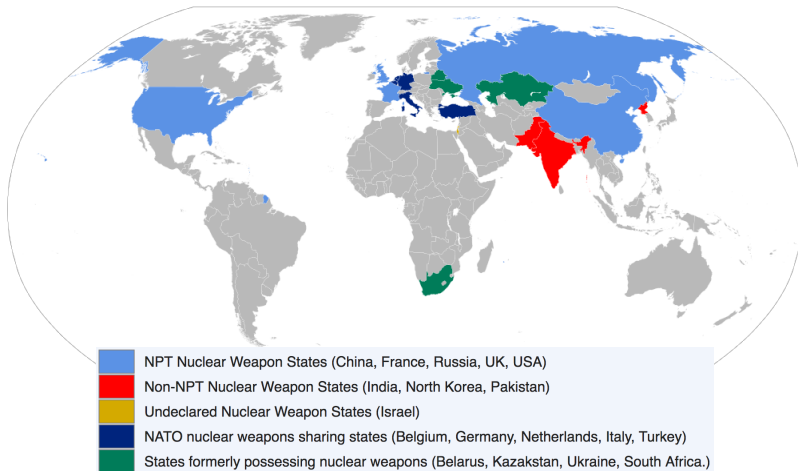
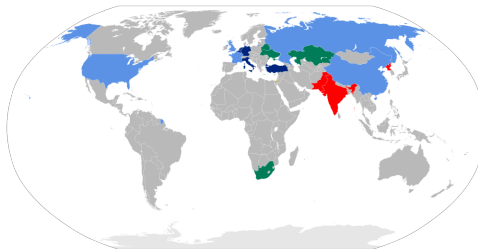
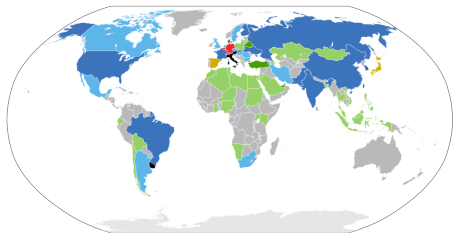


Figure 2: Nuclear weapons status of all nations [12].



Power vs. Weapons





International Reactors

Number of Power Reactors by Country and Status

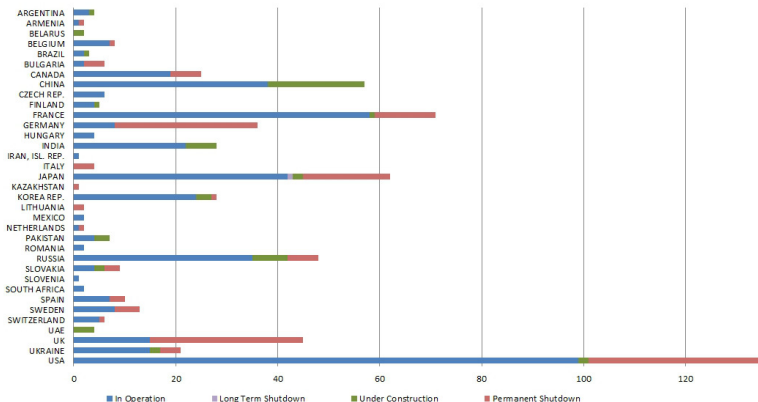


Figure 3: Nuclear reactors internationally, replicated from [8].



Nuclear Capacity

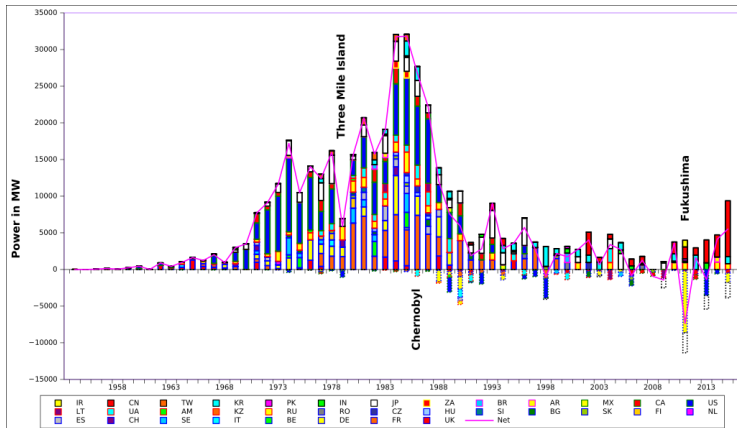


Figure 4: Nuclear power deployments as a function of time [15].



Spent Fuel Inventory

Radioactive Waste Volumes			
Type	In storage (m^3)	In disposal (m^3)	% in disposal
VLLW	2,356,000	7,906,000	77%
LLW	3,479,000	20,451,000	85%
ILW	460,000	107,000	19%
HLW	22,000	0	0%

Table 1: Solid radioactive waste volumes worldwide, IAEA estimate 2016. [?]



VLLW, LLW, ILW

Liquid

Must be solidified or, must be packed in absorbant package 2x liquid volume.
(i.e. decontamination solutions, liquid scintillators, ion-exchange fluids, etc.)

Wet Solid

Greater than 1% liquid, but primarily solid (i.e. filters).

Dry Solid

Less than 1% liquid (i.e. trash, swipes, clothes, tools, etc.)



Spent Fuel Inventory

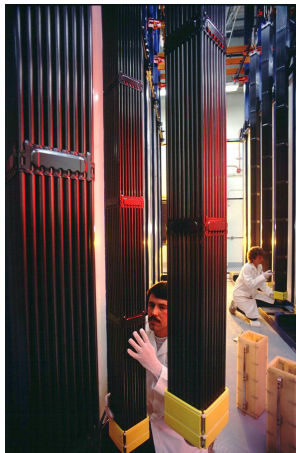


Figure 5: Spent nuclear fuel from conventional power reactors is in the form of uranium oxide fuel rods [?].



Spent Fuel

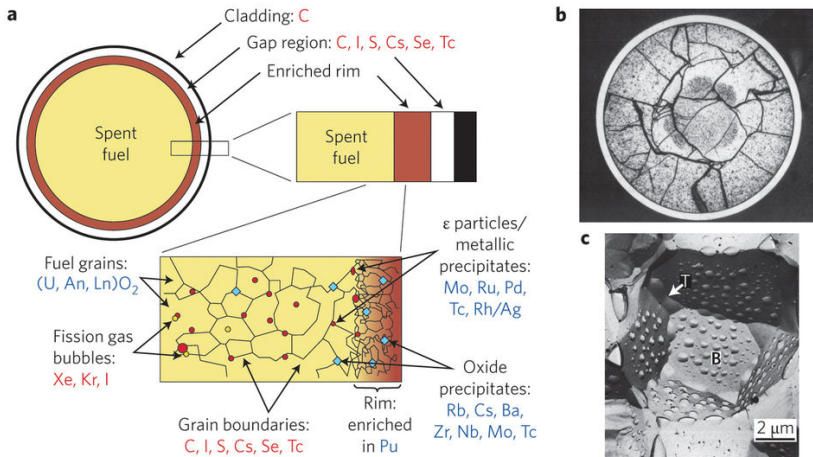


Figure 6: Microstructure of spent fuel and the distribution of fission products and actinides after irradiation in a reactor. From [?].



Spent Fuel Inventory

High Level Waste

- 300,000 metric tons worldwide [?]
- 90% in storage pools
- remainder in dry casks



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Long-Term Storage

Reprocessing

Deep Geologic Disposal

Array of Possible Options

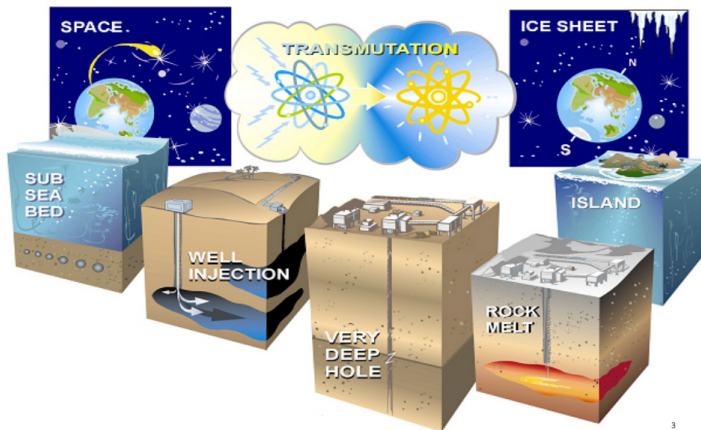


Figure 7: An array of options have been considered in the past [13].



VLLW, LLW, ILW

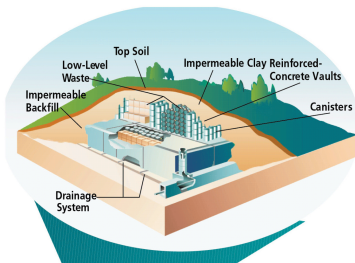


Figure 8: Design of a LLW repository.



Figure 9: Waste Control Specialists
Low Level Waste Repository in
Andrews County, Tx.



Spent Fuel Inventory



Figure 10: Spent fuel pools are at reactor sites and elsewhere [?].



Figure 11: Dry casks at reactor sites and elsewhere [10]



Reprocessing Waste

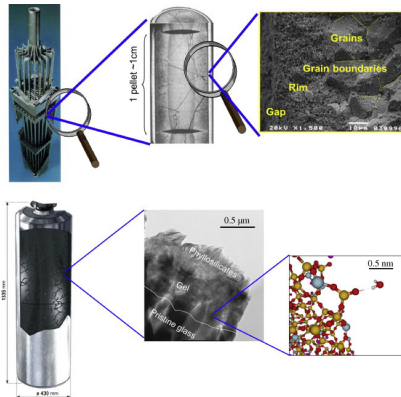


Figure 12: A comparison of uranium oxide and borosilicate glass waste forms [14].



Reprocessing Waste

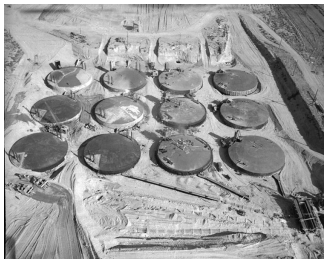


Figure 13: Liquid waste in steel or carbon steel tanks at Hanford and elsewhere[?].



Figure 14: Vitrified glass logs at reprocessing facilities and elsewhere [5].



Reprocessing Capacity

Global reprocessing capacity is shown in Fig. 15.

	Thermal Reactor UNF	Fast Reactor UNF
Research/Pilot/Demonstration Reprocessing Facility	Japan (Tokai facility) China (at Lanzhou) France (Atalante) India (BARC, IGARC) Italy (at Rotondella) Belgium (Eurochemic facility) Germany (WAK/Karlsruhe) Russia (Khoplin, Bochvar) United Kingdom (Sellafield) United States (national laboratories)	Russia France Japan United Kingdom United States (Argonne National Laboratory and Oak Ridge National Laboratory)
Commercial Reprocessing Facility	France (Marcoule and La Hague facilities) United Kingdom (THORP and Magnox reprocessing facilities at Sellafield) Russia (RT-I facility) United States (West Valley) India (Trombay, Tarapur, Kalpakkam)	France

Figure 15: [2].



MOX production

Global MOX production is shown in Fig. 16.

**TABLE II: WORLDWIDE MOX FUEL FABRICATION
CAPACITIES (T_{HM}/YR) IN 2009 AND 2015**

	2009	2015
France: MELOX	195	195
Japan: Tokai	10	10
Japan: Rokkasho	0	130
Russia: Mayak, Ozersk (pilot)	5	5
Russia: Zheleznogorsk (fast reactor fuel)	0	60
United Kingdom: Sellafield	40	0
Total for thermal reactors	250	400

Figure 16: [2].



Clay Disposal Environments

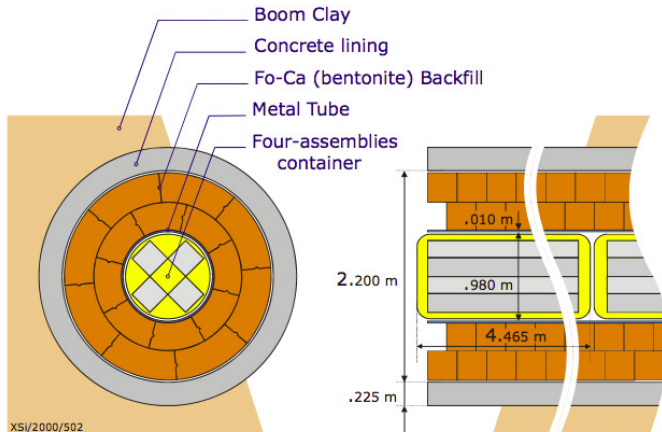


Figure 17: Belgian reference concept in Boom Clay [16].



Granite Disposal Environments

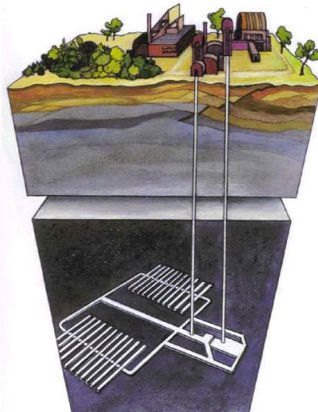


Figure 18: Czech reference concept in Granite [16].

Salt Disposal Environments

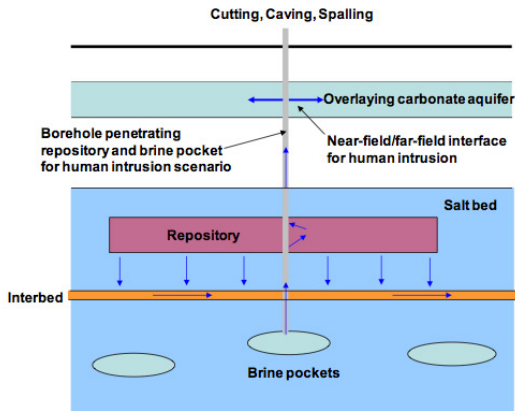


Figure 19: DOE-NE Used Fuel Disposition Campaign concept in Salt [4].



Deep Borehole Disposal Environment

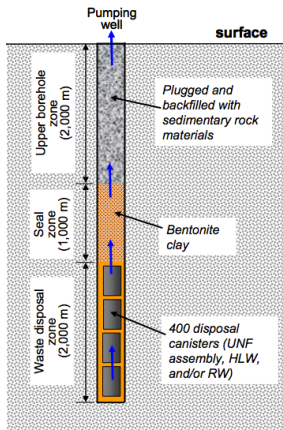
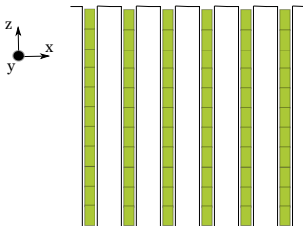


Figure 20: DOE-NE Used Fuel Disposition Campaign Deep Borehole concept [4].

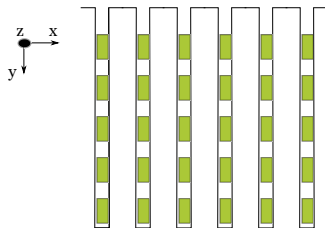


Repository Layouts

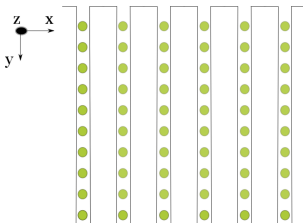
Deep Boreholes



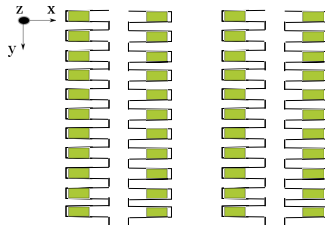
Horizontal In-Tunnel



Vertical In-Tunnel



Alcoves





All Disposal Environments

Features of Various Concepts

Feature	Clay	Granite	Salt	Deep Borehole
Hydrology				
Total Porosity [%]	34-60	0.1	0.5	0-0.5
Eff. Porosity [%]	0.5-5	0.0005	0.1	0.00005-0.01
Conductivity[m/s]	$10^{-11} - 10^{-9}$	$10^{-6} - 10^{-5}$	$10^{-12} - 10^{-10}$	$10^{-13} - 10^{-4}$
Fracturation	none	high	none	low at depth
Geochemistry				
Reducing	Near & Far Field	NF only	NF only	NF only
Oxidizing	none	Slight in FF	Slight in FF	Slight in FF
Salinity	higher at depth	higher at depth	high	high
pH	~ 7	≥ 7	≥ 7	~ 7
Design				
Waste Package	Steel, Cu	Steel, Cu	Steel	Steel, Cement
Buffer	-,Fo-Ca,Cement	Fo-Ca,Cement	Crushed Salt	-,Fo-Ca,Cement
Depth	100-500 m	100-500 m	100-500m	3-5km
Emplacement	Vert.,Horiz,Alcove	Vert.,Horiz.	Alcove	Vert.
Packages/Gallery	one, many	one, many	one, two	400
Thermal Behavior				
Buffer Limit [$^{\circ}\text{C}$]	100 (Fo-Ca)	100 (Fo-Ca)	180	100 (Fo-Ca)
Host Limit [$^{\circ}\text{C}$]	100 (alteration)	200 (cracking)	180 (brines)	none
Conductivity [$\frac{\text{W}}{\text{m}\cdot\text{K}}$]	1 - 2	2 - 4	~ 4	2 - 4
Coalescence	yes	no	yes	no



Repository Components

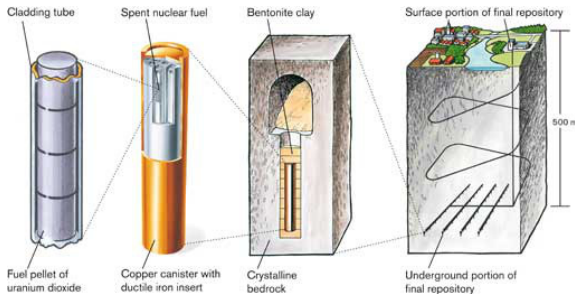
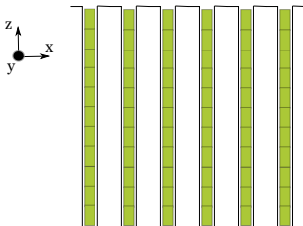


Figure 21: Geologic disposal systems typically employ engineered barrier systems as well as natural barrier systems. This is a Swedish concept in granite [1].

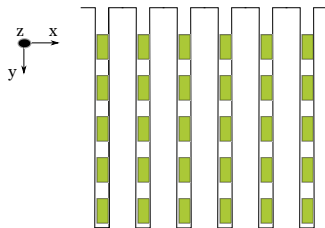


Repository Layouts

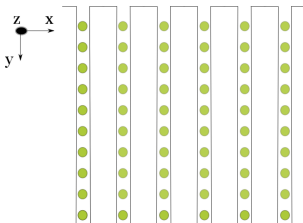
Deep Boreholes



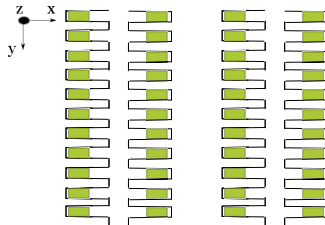
Horizontal In-Tunnel



Vertical In-Tunnel

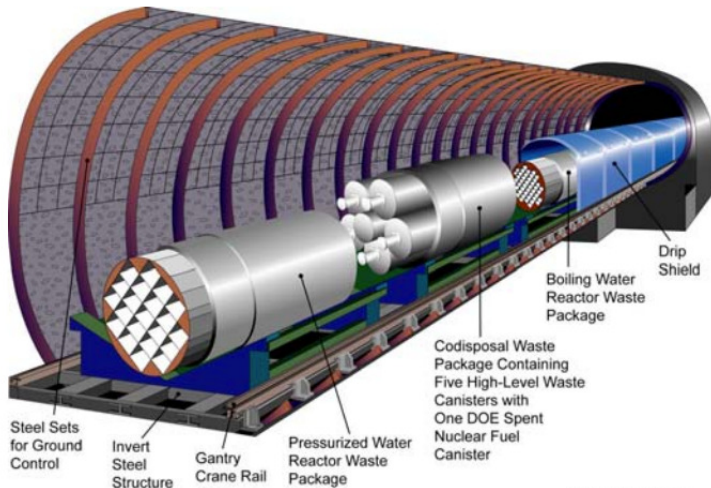


Alcoves





Unsaturated, Ventilated Concepts



Drawing Not to Scale
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Saturated , Enclosed Concepts

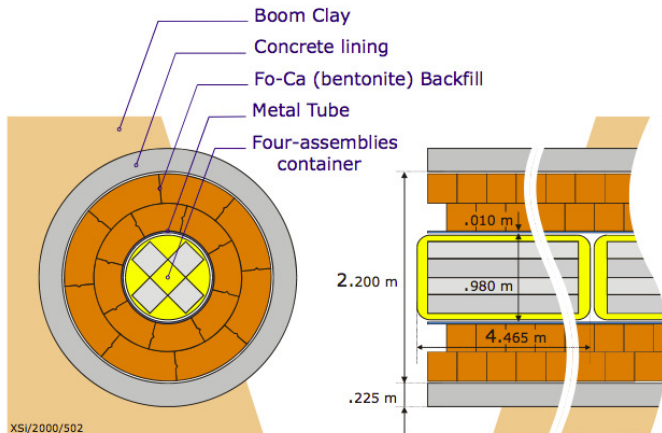


Figure 23: The Belgian reference concept in Boom Clay is backfilled very soon after waste emplacement without a ventilation period and is located below the water table [16].



Tuff (Yucca) Disposal Environments



Figure 24: Yucca Mountain is in southern Nevada [11].



Alternative Disposal Geology Options

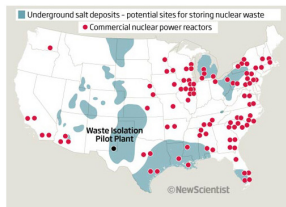


Figure 25: U.S. Salt Deposits, ref. [9].

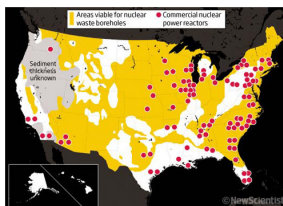


Figure 27: U.S. Crystalline Basement, ref. [9].

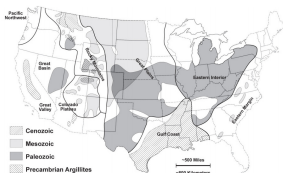


Figure 26: U.S. Clay Deposits, ref. [6].



Figure 28: U.S. Granite Beds, ref. [3].



Clay Disposal Environments

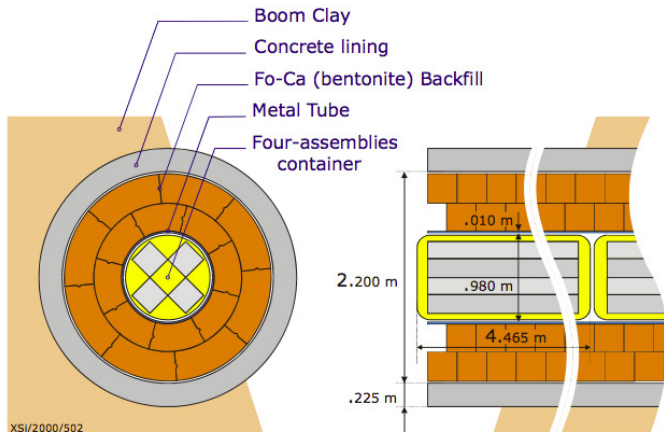


Figure 29: Belgian reference concept in Boom Clay [16].



Granite Disposal Environments

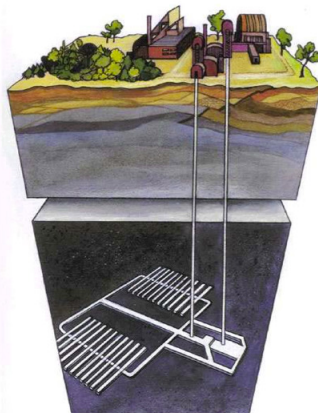


Figure 30: Czech reference concept in Granite [16].

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Figure 31: DOE-NE Used Fuel Disposition Campaign concept in Salt [7].



Salt Disposal Environments

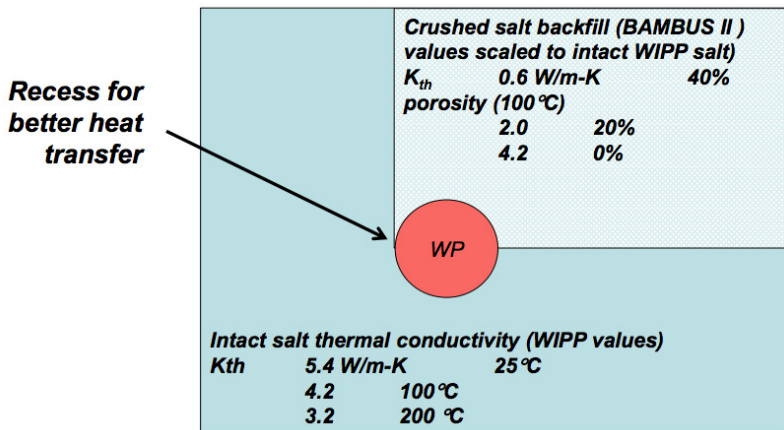


Figure 32: DOE-NE Used Fuel Disposition Campaign concept in Salt [7].



Deep Borehole Disposal Environment

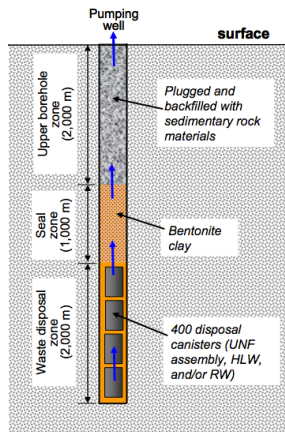


Figure 33: DOE-NE Used Fuel Disposition Campaign Deep Borehole concept [7].



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